

Unit 4, Lesson 4: Finding the Area of Composite Figures

Benchmark

MA.7.GR.1.2 Solve mathematical or real-world problems involving the areas of polygons or composite figures by decomposing them into triangles or quadrilaterals.

Learning Target

- ☐ I can decompose composite figures into triangles and quadrilaterals to find the area.

4.4.1 Warm-Up: Math Fail

The advertisement below is for the sale of colored strands for braids to raise money for a Stop Drug Abuse campaign.

**STREAKS
TO STOP
DRUG ABUSE**

1 STRAND FOR \$3
2 STANDS FOR \$5
3 STRANDS FOR \$9

WHAT?
I plan to bring a hint of red to everyone's head to raise awareness to the drug epidemic and support local drug addiction programs.

WHY?
Every strand of real red hair sold supports the teen-challenge program for young addicts and Plymouth New Hope addiction and protocols group.

HOW?
Simply ten minutes of your day will be used to put 100% really human rhinoceros red streaks to your real hair. These strands can be washed, styled, and dyed. Please see Mikaela Provola, Mrs. Darling, or Mrs. Baldner for further details.

**HELP ME TRY TO PUT AN
END TO DRUG ABUSE!**

Contact
Mikaela Provola

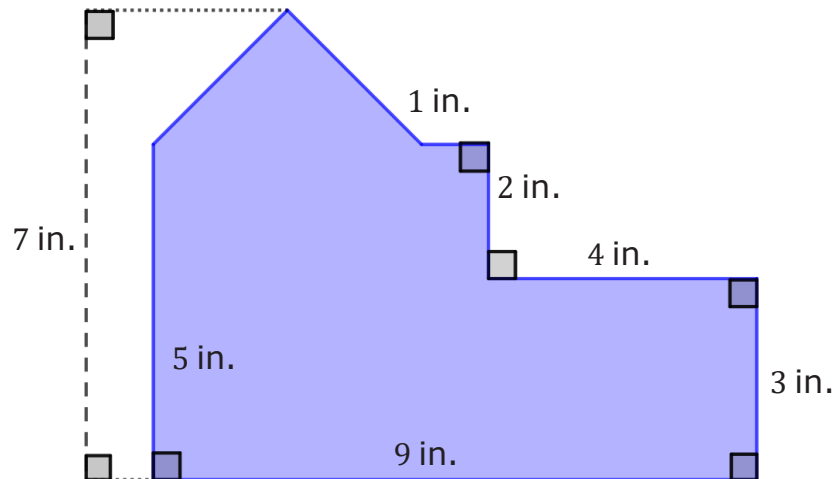
1. Explain what the math fail is in the advertisement.

4.4.2 Guided Instruction: Finding the Area of Polygons



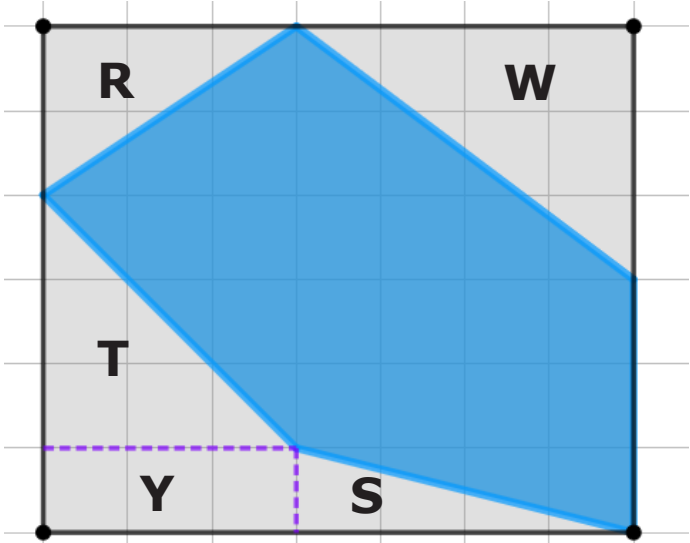
Composite figures are two- or three-dimensional figures that can be decomposed into smaller figures.

A floorplan is shown. All segment lengths are in inches (in.).

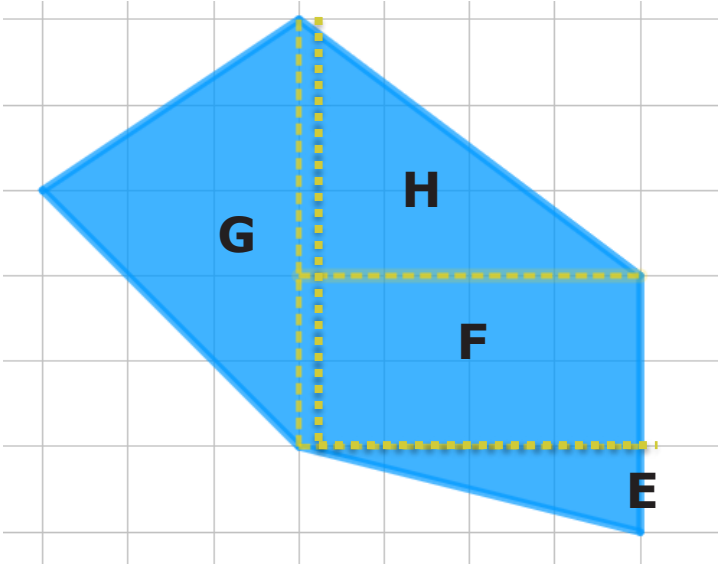


1. With your partner, decompose the floorplan into triangles and quadrilaterals.
2. Find the area of the floorplan. Show your work.
3. A polygon inscribed inside a rectangle is shown on the grid. Each grid line is 1 unit.

a. Complete the table.

Polygon Inscribed in a Rectangle	
	
Decomposition of Rectangle	
Area of the large rectangle	
Area of R	
Area of S	
Area of T	
Area of W	
Area of Y	
Area of the blue polygon	

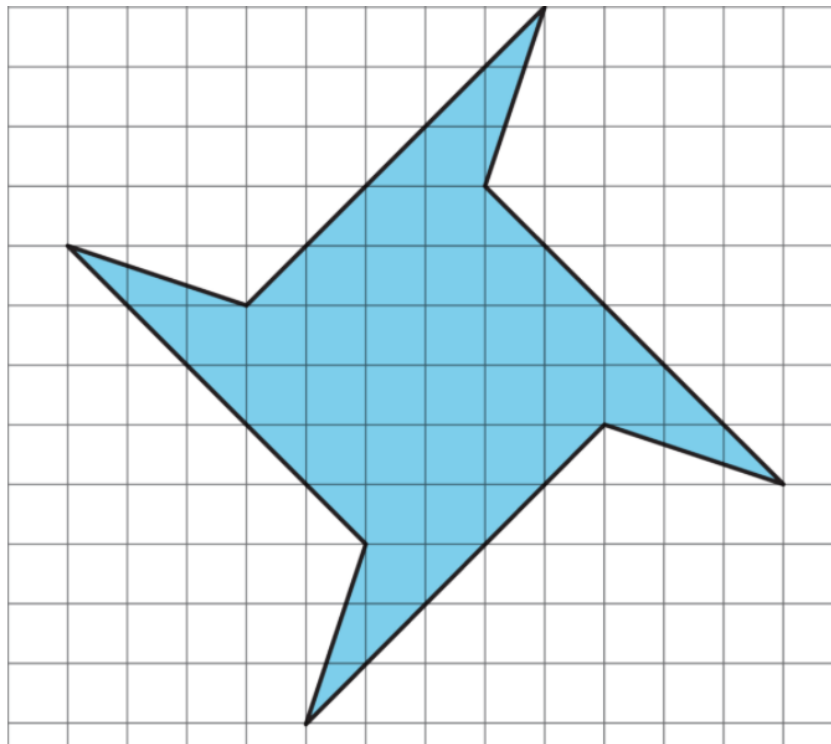
b. The same polygon is shown on the grid. Each grid line is 1 unit.

Polygon	
	
Decomposition of Polygon	
Area of E	
Area of F	
Area of G	
Area of H	
Area of the blue polygon	

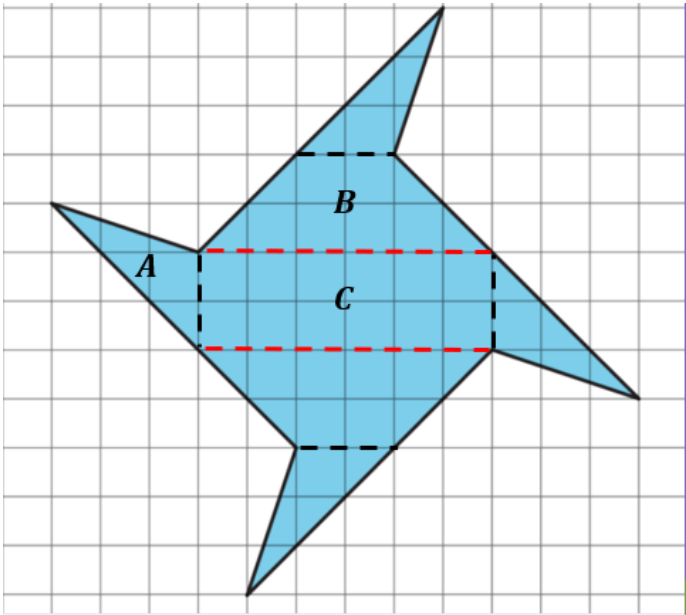
c. Discuss with your partner the different possible methods of finding the area of the blue polygon. List one strength of each method.

4.4.3 Collaboration: Area of a Polygon

1. A polygon is shown on grid paper. Each grid line is 1 unit.



- a. Decompose the polygon into triangles and quadrilaterals.
- b. Share your decomposition with your partner. List at least one similarity and one difference.
2. Taahira decomposed the polygon into 7 pieces using 3 different shapes, as shown. Each grid line is 1 unit.

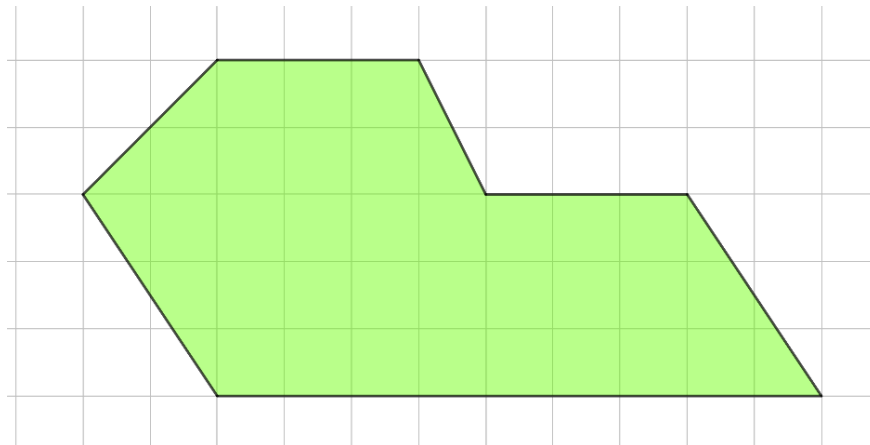


Work with your partner to find the area of each shape and the total area of the polygon.

Shape	Type of Shape	Area of the Shape	Number of Shapes	Total Area
A			4	
B				
C			1	
Area of the Polygon				

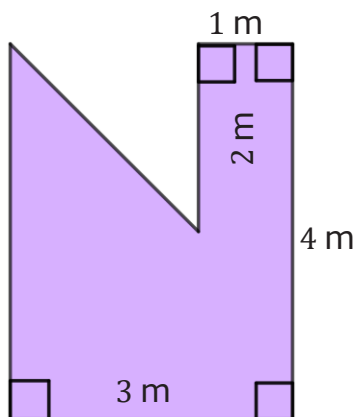
4.4.4 Guided Practice: Decomposing Composite Figures to Find Area

1. A polygon is shown on a grid. Each grid line is 1 unit.



Find the area of the polygon.

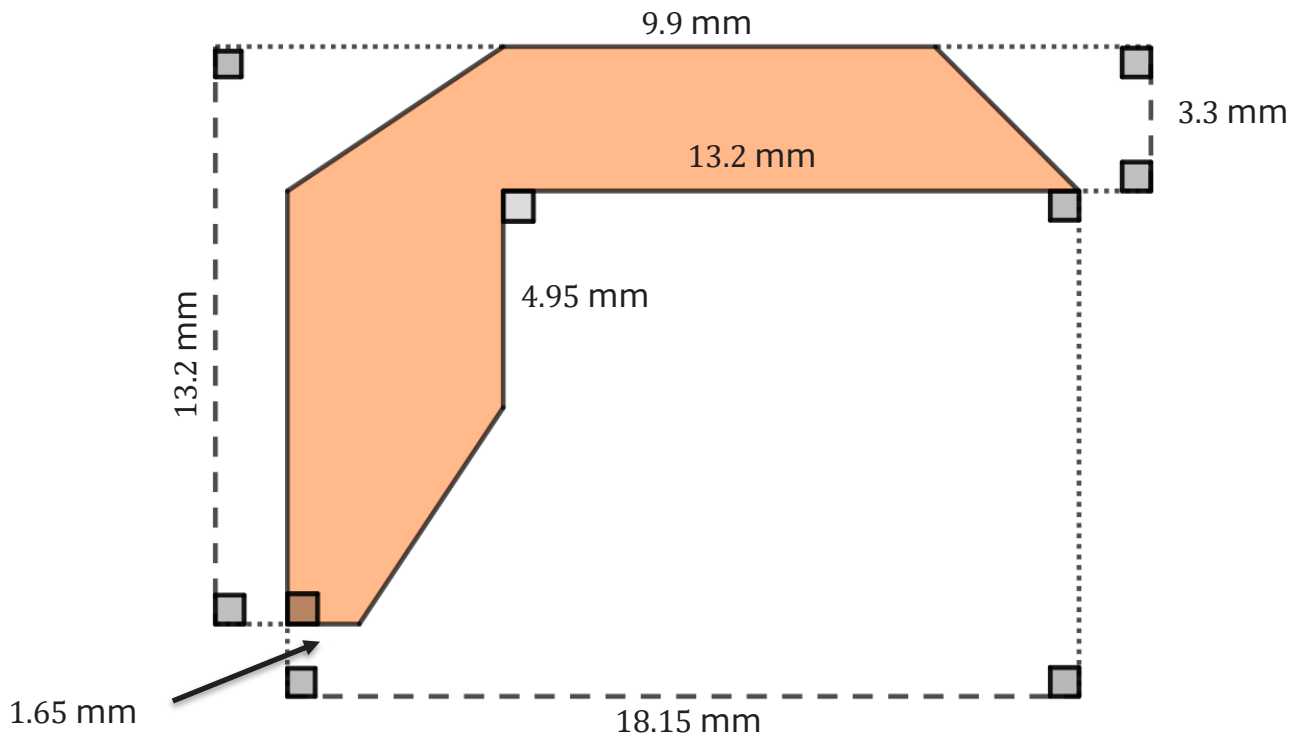
2. A polygon is shown with the given measurements in meters (m).



What is the area of the composite figure?

4.4.5 Your Turn!

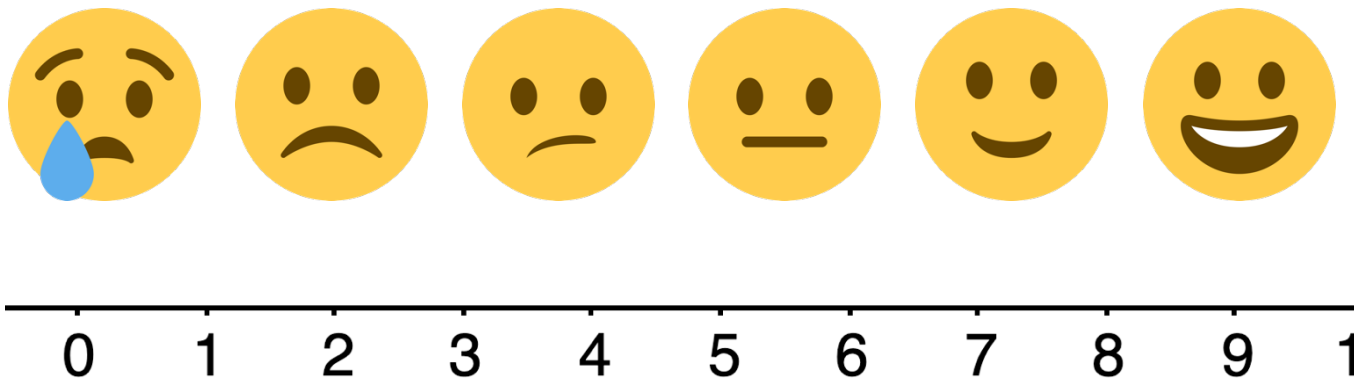
1. Find the area of the polygon. Lengths are shown in millimeters (mm).



4.4.6 Wrap-Up: Reflection

1. Place an **X** on the number in the emoji scale that indicates your understanding in relation to today's learning target.

I can decompose composite figures into triangles and quadrilaterals to find the area.



2. Provide evidence to explain your self-assessment.